Maddali Suhas Choudary

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Summary

Final-year B.Tech CSE student who builds industrial AI and safe agentic LLM systems end-to-end, from signal processing and model training to APIs, databases and CI. Built a vibration–acoustic predictive-maintenance pipeline validated on held-out bearings, and a security gateway that stops AI agents from executing harmful tool calls. Research intern at IIIT Dharwad on LoRA fine-tuning and LLM evaluation, where I showed standard metrics ranked models backwards. I report evaluation honestly, including baselines, confounds and negative results.

Education

Indian Institute of Information Technology, Senapati — Manipur, India
B.Tech in Computer Science and Engineering | CGPA: 7.62 — *2023 – 2027*

Technical Skills

Machine Learning: Python, PyTorch, HuggingFace Transformers, PEFT/LoRA, scikit-learn, NumPy, SciPy, pandas; fine-tuning (DeBERTa-v3, T5, GPT-2, Phi-3); grouped cross-validation, ROUGE-L, BLEU, BERTScore
Industrial AI & Signals: predictive maintenance, condition monitoring, anomaly detection, time-series, FFT and statistical features (RMS, kurtosis, crest factor), vibration and acoustic sensor fusion
GenAI & Agents: LangGraph, multi-agent systems, tool calling, structured outputs (Pydantic), MCP, agent security (prompt injection, tool poisoning), RAG, corrective RAG, hybrid retrieval (pgvector + full-text), hallucination detection
Engineering: FastAPI, REST APIs, SQL, PostgreSQL/pgvector, SQLite, Docker, GitHub Actions CI, pytest, Git

Experience

Research Intern — Dept. of Computer Science & Engineering — Jun 2026 – Aug 2026
Indian Institute of Information Technology Dharwad | Supervisor: Dr. Krishnendu Ghosh — *Dharwad, India*

- Built an end-to-end curriculum question-generation system: a **FastAPI** backend serving five LLMs (T5, GPT-2, LLaMA 3.2, Gemma, Phi-3) behind a React UI, a pipeline mining (context, question) pairs from NCERT textbooks, and **LoRA/PEFT fine-tuning**; evaluated **1,400 generated questions** across 25 class-subject pairs, with tuning improving 74% of metric scores (636 of 864).
- Showed that **reference-based metrics ranked models in reverse order of real quality** (one base model hit 0.96 ROUGE-L by copying a PDF footer) and designed a keyword-grounding relevance and hallucination metric that fixed the ranking; fine-tuning lifted the grounded-question rate from **3.2% to 91.2%** (GPT-2) and 58.9% to 86.4% (Phi-3).

Projects

FabGuard AI (Bearing Condition Monitoring) | *scikit-learn, LangGraph, PostgreSQL* | GitHub — Sep 2026 – Present

- Built a fault-detection pipeline on 20 bearings with paired 42 kHz **vibration and acoustic** signals: FFT band-power, RMS, kurtosis and crest-factor features, with models compared under **nested leave-one-bearing-out cross-validation**.
- Shipped a **fusion Isolation Forest trained only on healthy data**: **0.955 balanced accuracy** and **98.9% developing-fault recall** at 5.6% false positives over 15 held-out bearings, 8.6 ms per window on CPU; a confound audit exposed a load shortcut, so 5 affected bearings were excluded.
- Engineered a **human-in-the-loop LangGraph agent** over hybrid pgvector + full-text retrieval; a deterministic verifier fails closed on any report whose numbers differ from the model's outputs or that cites unreviewed or injection-shaped passages. **PostgreSQL checkpointing resumes after a worker crash**; role-gated FastAPI, idempotent approvals, 68 tests in GitHub Actions CI.

IntentShield (Guardrail for AI Agents) | *Python, MCP, FastAPI, SQLite* | GitHub — Sep 2026 – Present

- Built an offline-first **security gateway for tool-using AI agents** that judges every MCP tool call as ALLOW, REVIEW or BLOCK before any side effect, via hooks for Claude Code, Codex CLI, Cursor and Gemini CLI plus a transparent MCP proxy: deterministic policy with reason codes, **tool-poisoning and rug-pull detection**, output taint tracking that blocks attacker-supplied arguments, and human approval bound to the exact call and an expiry.
- On **220 held-out MSB attacks** against live MCP servers, cut executed harm from **58.5% to 0.0%**; a trivial baseline I added also hit 0%, so I measured the real gain: **0 harmful calls left for human review versus 47** for the baseline. Documented negative results too (a DeBERTa injection classifier flagged 24% of benign text).

Hallu-Check (Hallucination Detection) | *PyTorch, FastAPI, DeBERTa-v3* | GitHub — Jan 2026 – May 2026

- Built an **8-node agentic pipeline** with a two-stage router (regex rules, LLM fallback) and **corrective RAG** that grades each scraped chunk with an LLM judge, and rewrites the query when nothing is relevant.
- Fine-tuned **DeBERTa-v3-base on 392K MNLI pairs** as a local claim verifier (about 42 ms per claim); claims route to NLI, SymPy, or LLM-written unit tests in a sandboxed subprocess, then up to 3 refine-and-reverify rounds.